Computer Vision – 21CSE454T

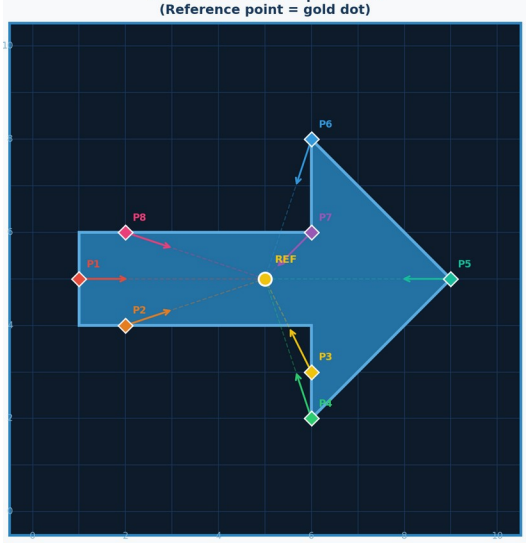
Unit 3 - Session 6

SLO 2: Generalized Hough Transform (GHT)- Design shape templates for GHT voting systems

Duration: 10 Minutes Activity

Question:

PART A — Arrow Shape on Grid
(Reference point = gold dot)



PART B — Fill in the R-Table

For each boundary point: compute r and phi (phi = angle of vector from Pn to REF)

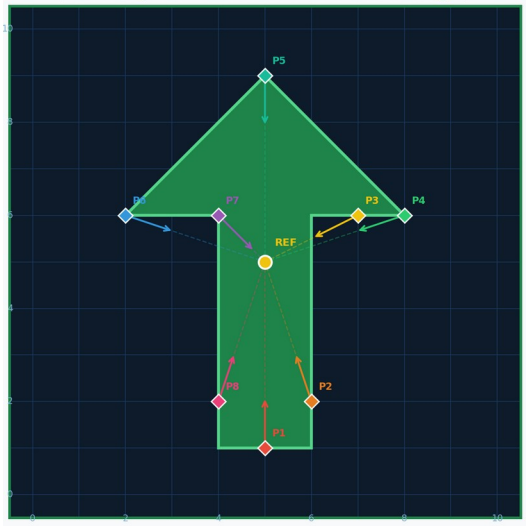
Point	Boundary (x, y)	Vector to REF (dx, dy)	r = sqrt(dx²+dy²)	phi = atan2(dy,dx) (degrees)
P1	(1, 5)	(4, 0)	4.00	0.00°
P2	(2, 4)	(3, 1)	3.16	18.43°
P3	(6, 3)	(-1, 2)	2.24	116.57°
P4	(6, 2)	(-1, 3)	3.16	108.43°
P5	(9, 5)	(-4, 0)	4.00	180.00°
P6	(6, 8)	(-1, -3)	3.16	251.57°
P7	(6, 6)	(-1, -1)	1.41	225.00°
P8	(2, 6)	(3, -1)	3.16	341.57°

FORMULAS

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dx = REF_x - P_x          dy = REF_y - P_y
r = sqrt(dx^2 + dy^2)     phi = atan2(dy, dx)  [convert to degrees]
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REF = (5, 5) | phi range: use 0 deg to 360 deg (add 360 if negative)

PART C — Partner's Shape (Rotated 90 deg)
(Same REF point — how do the R-table entries change?)



PART D — Partner's R-Table (Rotated Shape)

Fill in r and phi for the rotated boundary points below

Point	Rotated (x, y)	Vector to REF (dx, dy)	r	phi (deg)
P1	(5.0, 1.0)	(0, 4)	4.00	90.00°
P2	(6.0, 2.0)	(-1, 3)	3.16	108.43°
P3	(7.0, 6.0)	(-2, -1)	2.24	206.57°
P4	(8.0, 6.0)	(-3, -1)	3.16	198.43°
P5	(5.0, 9.0)	(0, -4)	4.00	270.00°
P6	(2.0, 6.0)	(3, -1)	3.16	341.57°
P7	(4.0, 6.0)	(1, -1)	1.41	315.00°
P8	(4.0, 2.0)	(1, 3)	3.16	71.57°

COMPARE WITH PART B

i) What happened to the r values after rotation?

The r values remained unchanged.

ii) What happened to the phi values after rotation?

The phi values increased by 90° (mod 360°).

iii) Why does GHT need a separate R-table per orientation?

Because the R-table depends on the shape orientation.